CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student brings the north pole of a bar magnet near the north end of a compass needle. The needle's north end will: 1

(a) stay unchanged **(b)** move closer **(c)** turn away **(d)** melt

2. Which action would not demagnetise a bar magnet? 1

(a) heating it red hot **(b)** hammering it

(c) keeping it with a keeper **(d)** dropping repeatedly

3. The strength of a compass reading is best when it is placed: 1

(a) near speakers **(b)** near a fridge

(c) away from iron and magnets **(d)** near a steel desk

4. Which one of the following sets lists items that all can be stuck to a fridge magnet? 1

(a) plastic cap, wooden peg **(b)** copper coin, brass key

(c) glass bead, rubber **(d)** steel clip, iron nail

5. If the poles of a magnet are painted, how would you check the marking? 1

(a) suspend it and see which end points north

(b) taste it

(c) weigh it

(d) look at colour

6. Which of the following devices works because of magnets? 1

(a) a plastic pen **(b)** a loudspeaker **(c)** a candle **(d)** a spoon

7. A magnet is cut into three equal pieces. The number of magnets obtained is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 6

8. A bar attracts the north end of a compass needle at one end and the south end of the needle at the other end. The bar is most likely: 1

(a) unmagnetised iron **(b)** wood

(c) a magnet **(d)** plastic

9. Which statement about a bar magnet suspended in a gap between two other magnets is correct? 1

(a) it stays still **(b)** it melts

(c) it flies **(d)** it turns to align with the field

10. Which of the following decreases the strength of the force between two magnets? 1

(a) using stronger magnets **(b)** increasing the distance
(c) bringing N-S closer **(d)** decreasing the distance

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Give reasons why a magnet is kept on the tabletop away from iron objects when not used. 2

12. Explain why magnetic poles always occur in pairs. 2

13. Why is a ceramic or plastic container used for storing magnets in a lab? 2

14. Why can a magnet pick a pin from a pile faster than hands? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. If you drop a magnet onto the floor repeatedly, what will happen and why? 3

16. How are magnets useful in the kitchen? 3

17. How can you demonstrate that magnets lose power on heating? 3

OR

How will you know if a cord of iron wire is magnetised or not? 3

18. What happens when a very strong magnet is brought near a compass? 3

OR

Explain why a hanging magnet turns slowly. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A student says: 'Magnet attracts all metals, so it can be used to separate gold from sand.' Evaluate and suggest the right method. 5

OR

Describe how to distinguish between poles of a magnet using a magnetic needle, with a diagram in words. 5

20. Explain how the compass was an important invention and how its working relates to the Earth's magnetism. 5

OR

Explain the following with examples: temporary magnet, permanent magnet, natural magnet. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND